Project Atlas — Architecture Requirements

Project Overview

Project Atlas is an internal analytics platform designed to process transactional data and provide near-real-time business reporting.

Technical Requirements

- Data ingestion: 350 GB of transactional data daily.
- Data residency: All customer data must remain within the European Union.
- Reporting latency: Maximum acceptable delay of 15 minutes.
- Cloud budget: $4,000 per month.
- Infrastructure: Preference for fully managed cloud services.
- Security: Role-based access control (RBAC) is mandatory.

Additional Considerations

The architecture should support future growth in data volumes, provide high availability, and minimize operational overhead.

The technology stack has not yet been selected. BigQuery, Snowflake, and Databricks are potential candidates.

Open Questions

- Which cloud data platform best satisfies these requirements?
- What are the expected monthly storage and processing costs?
- How should the solution handle data ingestion, transformation, and reporting?